Fernando Jr. Villanueva

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https://fvjr156.github.io/my-portfolio

Passionate software developer with freelancing experience in crafting high-performance mobile applications using Java, Android Studio, and Room Databases, alongside scalable web applications powered by React, Express, MySQL, and Node.js.

Skills

- Backend Web Development
- Frontend Web Development
- Mobile App Development
- Research and Technical Documentation
- MySQL, MongoDB, SQL Server, SQLite, Room
- Git and GitHub
- Tailwind CSS and Bootstrap
- HTML and CSS Proficiency
- JavaScript and TypeScript Proficiency
- Java and C# Proficiency
- C/C++ and Low-Level Programming Familiarity
- MERN stack and MVC architecture familiarity

Work Experience

ML-Based Sign Language Translator Application
Contract Developer — May to December 2025

- Developed an Android application which uses a TFLite machine learning model to translate certain American Sign Language gestures into text and speech.
- Built a user-friendly and easy to navigate User Interface to allow users to quickly access different pages for translation, settings, and sign language gesture references.

Augmented Reality PC Assembly and Disassembly Application
Contract Developer — May to August 2025

- Developed an Android application which uses Augmented Reality and 3D models to simulate PC assembly and disassembly processes.
- Implemented marker recognition for AR model spawning and rendering using Unity and Vuforia.

Education

Bachelor of Science in Computer Science
STI College Marikina
Expected finish March 2028

Years 1-2:

- Developed multi-platform applications using ASP.NET (web/mobile/desktop), Android Studio (Java), and Java AWT, targeting diverse devices and operating systems.
- Managed SQL Server databases using SQL Server and SQL Server Management Studio, with advanced queries, stored procedures, and database normalization.
- Implemented data structures (e.g. arrays, trees, graphs) and algorithms. Designed and analyzed efficient algorithms for real-world scenarios.
- Applied Object-Oriented Programming (OOP), Functional Programming, and MVC architecture in coding exercises and hands-

on projects, enhancing code modularity, scalability, and maintainability.

SHS — IT in Mobile App and Web Development Strand
STI College Marikina
August 2022 – July 2024

- Mastered 3 core programming languages (Java, C#, JavaScript) through hands-on projects and lessons, building foundational coding proficiency.
- Developed interactive web applications using the web technologies HTML, CSS, and JavaScript, implementing responsive web designs and smooth client-side functionality.
- Applied Object-Oriented Programming (OOP) and Functional programming concepts in hands-on projects, lessons, and coding exercises, optimizing code structure, modularity, scalability, and maintainability.
- Built one (1) mobile application using Android Studio and Java, and one (1) full-stack web-based application using MERN stack and MVC architecture, as parts of major course deliverables.
- Designed and queried relational databases using SQL Server and SQL Server Management Studio, executing CRUD operations and complex joins for data-driven projects.

Internship

Data Encoder Intern — Student-Trainee
National Bureau of Investigation, Marikina City Satellite Office
February to March 2024 (1 month)

- Requirement for SHS Work Immersion course
- Responsibilities:
  - Processed data for 500+ NBI clearance applicants (up to 50 applicants per day), ensuring 100% accuracy in record digitization.
  - Verified identities of 500+ applicants through submitted legal document cross-checks (e.g. birth certificates and past clearance applications), upholding data entry accuracy.
  - Guided 300+ applicants through clearance application processes, streamlining application submissions and resolving queries to improve turnaround times.
  - Contributed to a positive work environment by proactively assisting colleagues as needed, fostering strong professional relationships and a collaborative atmosphere.
  - Developed strong attention-to-detail skills through the constant review of entered information, ensuring high-quality results.
  - Used computer software to encode, store, and retrieve data.
  - Maintained strict and utmost confidentiality while handling sensitive client information according to company guidelines and legal regulations.

Certifications

- **Crash Course on Python**
  issued by Coursera — April 2026
  https://coursera.org/share/4d166512610af4021eee04ae9c640856

- **Foundations of Agile Project Management**
  issued by HackerRank — March 2026
  https://coursera.org/share/2592eec1c05cf07d832f044d8ff2de6f
- **React (Basic)**
  issued by HackerRank — November 2024
  https://www.hackerrank.com/certificates/6e931408ac0a
- **Java (Basic)**
  issued by HackerRank — November 2024
  https://www.hackerrank.com/certificates/19689f072134

Languages

- **English** – Fluent, Working Proficiency
- **Filipino (Tagalog)** - Native